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Case Study: Genetic Privacy

Answer to the question number 1

Genetic privacy mainly affects the society and it does not affect the environment directly as I can see. So far it can give personalized medicine to humans and diagnose diseases early. It does have some potential for the future. If a new disease is created then it will do more harm than good and will cause problems for us.

If it works successfully then there will not be any long term risk which is unlikely to happen. Once people thought that smoking was good for health but after 50 years they got enough dead bodies to figure out that it wasnt, same here we can not say its long term effect with little data.

These are really bad for our environment like GPT-3 emitting ~500 metric tons of CO_2, so keeping them in the long term will be bad for us. As they could only give us information but they will take all our resources from our environment.

It creates a "genetic underclass" they cannot benefit from the system.

Answer to the question number 2

I don't think the system has ethical violations because Genetic Privacy is something that does not need any privacy at all. I mean why do you need any privacy on this matter at all? What bad will it do if someone knows what my eye colour is or what my height, this are useless things to worry about, and privacy is definitely not needed here.

But the trained system will be biased, Most genomic datasets used to train AI models are overwhelmingly with European (often cited as >80% European ancestry). This is known as the "ancestral bias" in genomics. European people will have advantage but others will suffer, this will only be a problem if and only if the model is weak but most of the hyper advanced models will not have any problem.

AI and Transparency doesn't go hand in hand, AI is like a black box, Transparency doesnt goes with it. We don't even fully understand how an AI works.

Is the system/company holds itself accountable for any violations, it comes down how we define our model to be. If we design it to be accountable for its mistake then it will be but it doesn't matter if it is accountable or not, what good will it do? Being accountable for its mistake wont make its mistake go back nor it could learn from its mistakes like humans do.

Answer to the question number 3

It can be used to deny life insurance to any human if they have disease prediction by AI for the future. It is good for insurance companies but bad for humans overall. If there is a new law created for this then everyone can get its benefit, again not a big problem.

Engineers must implement Privacy when designing it. This means using techniques like Differential Privacy or Federated Learning or other. Even after release of the AI the model should be kept under test if that is still working without any problem. But Engineers of this caliber are rare and hard to find both good in Engineers and in the medical field to then need a big team to run this. Legal this wouldn't be anyone’s head if Genetic Privacy would be considered as a privacy to begin with.

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